

Point sur les méthodes AD/AI (MV)

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Full AI

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- It implies that $\mathbb{P}(\max(X) > u) = \mathbb{P}(X_1 > u) + \mathbb{P}(X_2 > u)$ for $u \rightarrow +\infty$.

Simulation of extreme events in the full AI case

- Denoting $P_1(x, y) = \mathbb{P}(X_1 \leq x, X_2 \leq y \mid X_1 > u)$ and $P_2(x, y) = \mathbb{P}(X_1 \leq x, X_2 \leq y \mid X_2 > u)$ the two conditional distributions.

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- Secondly, for $(x, y) / \max(x, y) > u$, we end up with

$$\mathbb{P}(X_1 \leq x, X_2 \leq y \mid \max(X) > u) = w_1 P_1(x, y) + w_2 P_2(x, y)$$

with $w_1 = \mathbb{P}(X_1 > u \mid \max(X) > u)$, $w_2 = \mathbb{P}(X_2 > u \mid \max(X) > u)$

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- The weights w_1, w_2 represents the weights of the different scenarios.

Behaviour of each coordinate for extreme observations

- We can decompose $P_1(x, y)$ into two quantities:

$$P_1(x, y) = \mathbb{P}(X_1 \leq x \mid X_1 > u) \mathbb{P}(X_2 \leq y \mid u < X \leq x)$$

- We identify an **exceedance law** of one marginal and a **conditional law**.
- We can use this latter to describe what is the tail behaviour of the vector.

- Let show that $\mathbb{P}(X_2 \leq y \mid u < X_1 \leq x)$ will equal 1 if $(y > u)$ and $(x > u)$.

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- Since $\mathbb{P}(X_2 > y, X_1 \in (u, x]) = \mathbb{P}(X_2 > y, X_1 > u) - \mathbb{P}(X_2 > y, X_1 > x)$, we obtain

$$\mathbb{P}(X_2 > y, X_1 \in (u, x]) < \mathbb{P}(X_1 > u, X_2 > u) \rightarrow 0$$

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Behaviour of the conditional law

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- Thus, $\mathbb{P}(X_2 < y \mid u < X_1 \leq x) \rightarrow 1$ (complement of the event) when $\min(x, y) > u$ and $u \rightarrow +\infty$

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- It implies that when the first coordinate is above the threshold ($x > u$), the second one is strictly below it.
- Same idea for the case $X_2 > u$. **Only one** coordinate is above the threshold u for each realisation of the law.

Full and partial AD case

- $\chi_{1,2}(u) \not\rightarrow 0$ implies that $\mathbb{P}(\max(X) > u) \neq \mathbb{P}(X_2 > u) + \mathbb{P}(X_1 > u)$ when $u \rightarrow +\infty$.
- Bivariate extreme values model (bev) can be used to approach such distributions.

- It is assumed here that extreme levels of $\min(X, Y)$ can be reached (Legrand 2022).
- Using a polar coordinate representation, we simulate new angles with a parametric (Tawn 1988) or a non-parametric method (Legrand 2022).

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- Following (Mourahib, Kiriliouk, and Segers 2025), let $J_k \subset [[1, d]]$ and $C_k = \{X_j > u \ \forall j \in J_k, \ X_{j'} < u \ \forall j' \in [[1, d]] \setminus J_k\}$. C_k corresponds to an **extreme direction**.

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- We get

$$\mathbb{P}(\max(X) > u) = \sum_{k=1}^K \mathbb{P}(X \in C_k)$$

where $1 < K < d$ is the number of scenarios.

- If $K = d$, we retrieve the AI case. If $K = 1$, we retrieve the full AD structure.

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- For each $k = 1, \dots, K$, a mev model is used on the subvector X_{J_k} whereas other coordinates are below u . If we use the model of (Mourahib, Kiriliouk, and Segers 2025), these latter equal 0 (lower bound with a Frechet scale).